

A Novel Robust Adaptive Impedance Control Scheme for Contact Force Tracking With Unknown Environment

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Abstract—Tracking desired force on various environmental surfaces is essential in many modern robotic applications. The constant impedance control (CIPC) has ability to realize this goal with clear environmental information (including stiffness and geometry), but failed when the information is not available. The traditional adaptive impedance control (TAIPC) is capable of tracking desired force even when the stiffness and geometry of environment are unknown, but large force overshoot is caused in contact phase. To cope with drawbacks existed in above two methods, a novel brief and effective robust adaptive impedance control scheme is proposed in this work, which has good ability for desired force tracking with unknown environmental information. A new adaptive compensator is introduced into the impedance framework to deal with lumped uncertainties caused by unknown stiffness and geometry of environment. It can minimize force tracking error and the force overshoot has also been reduced a lot. The stability and convergence analysis is conducted using a spring model, but the method is not applicable to a complex second-order environmental model. Simulation and experimental results sufficiently demonstrate the effectiveness and confirm the theoretical claims of the method.

Index Terms—Robot force control, contact force tracking, adaptive impedance control, unknown environment.

I. INTRODUCTION

CONTACT force control has important applications in modern robotics. Typical force control scenarios include grinding, polishing and assembly, etc [1], [2]. In such scenarios, the contact force between robot manipulator and environment needs to be controlled with desired values to complete specific tasks. High force tracking accuracy is easy to be achieved when the environmental stiffness and geometry are known. However, environmental information is usually partly known or unknown in practical applications, which leads to the high-precision force control a challenging task [3].

In the past few decades, many scholars put their attentions on the force tracking problem. The main research results include hybrid position/force control [4] and impedance control [5].

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Ding et al. [6] proposed a task-oriented adaptive position/force control scheme with hybrid constraints. A reference trajectory generation method is employed to obtain and shape the command trajectory that guarantees the safety contact. Abbas et al. [7] designed an event-triggered adaptive hybrid position/force control for networked ultrasound robotic manipulator. The event-triggered mechanism is applied to the force control to update the control inputs in an aperiodic manner, reduce the communication burden and improve force tracking performance. On the other hand, after the core work of impedance control has been proposed by Hogan [5], many extended researches have been developed to improve the robustness and tracking accuracy of the original impedance function. Gu et al. [8] proposed a fractional-order PID impedance control for contact force tracking for manipulators, the fractional order integration and differentiation technique have been employed to enhance system robustness. Zhou et al. [9] proposed a robust sliding mode impedance control for complex force controlled scenarios. A new impedance control model is proposed and a disturbance observer is designed to cope with environmental and dynamic uncertainties. Jin et al. [10] designed a model predictive impedance control scheme with one-step and multistep model predictive optimization, the impedance law design problem has been transformed into a control law design problem. Both hybrid position/force control and conventional constant impedance control can deliver effective contact force tracking when operating on plane surfaces. However, when the operating environments are slope surfaces or more complex curved surfaces, force controlling accuracy can not be guaranteed. This is because that the stiffness and geometry of complex environment are not easy to be obtained [11], which greatly reduces the control performance in traditional force control methods.

It is common for robot manipulators to perform force control operations in complex environments. Due to the unknown stiffness and geometry of environments, accurate force tracking becomes a challenging task. To cope with environmental uncertainties, many variable impedance control schemes have been proposed, such as fuzzy impedance control, neural network impedance control, optimization base impedance control and adaptive impedance control, etc. These methods aim to adjust parameters of impedance function online to achieve a better force control result in different environments [12], [13], [14], although the environmental information is partly known or unknown. Liu et al. [15] designed a fuzzy logic controller to

modify damping parameters online to adapt to changes in the environment, which achieved a good force tracking result on complex surfaces. Hamedani et al. [16] proposed an intelligent force tracking control strategy using Wavelet Neural Network (WNN). The WNN is employed to compensate environmental uncertainties caused by unknown stiffness and geometry to minimize force tracking error. Jin et al. [17] proposed an optimal variable impedance control method, which transforms the force control problem with uncertainties to an optimal law design problem. The force error has been minimized with generated optimal impedance parameters. But fuzzy logic and neural network need large calculation, optimal control needs to iteratively solve an optimization problem per control cycle. They are not easy to implement in application scenarios with very fast control period. By contrast, adaptive impedance control (AIPC) is easy to realize and has good robustness to unknown environment. Since it is proposed by Seul et al. [18], the AIPC has been widely studied and used in many works. In [19], Duan et al. developed an adaptive variable impedance control method to ensure accurate force tracking in environments with unknown characteristics. In [20], Dong et al. designed an adaptive variable impedance control scheme that incorporates a method for extracting the actual interaction force. The experiments on TX-90 robot verified its good force tracking result. In [21], Dong et al. presented a dual closed-loop adaptive decentralized control framework aimed at improving physical human-robot interaction ability. The AIPC has been combined with neural network to achieve dynamic force tracking in uncertain interaction situations. In [22], Wang et al. designed a PD-adaptive variable impedance force control for compliant grinding and polishing. A PD control has been added into AIPC framework to achieve fast force response speed and reduce force overshoot. However, above works are based on the traditional adaptive impedance control (TAIPC) framework proposed in [18], which can minimize force error under unknown environment, but large force overshoot will be caused in contact phase. It is a main drawback in TAIPC.

In this work, we proposed a novel robust adaptive impedance control for contact force tracking toward unknown environment. The controller is constructed with a new adaptive law and a PD control term. The PD control can be used to handle uncertainties, as shown in works [8], [15], [22]. But only a PD control can not achieve high-precision force tracking and noticeable force overshoot reduction. Therefore, we introduced a new adaptive law, which uses the information of force tracking error and control input to update itself and then provide a feedforward compensation for uncertainties. More importantly, the proposed adaptive compensator can provide a designable initial compensation during contact transition, this can significantly reduce force overshoot compared with PD controller and existing AIPC methods [18], [19], [20], [21], [22]. Therefore, high-precision force tracking and noticeable force overshoot reduction can be achieved with the proposed adaptive compensator. The stability analysis of the method can be guaranteed with a spring model, but it is not suitable for a complex second-order environmental model. For the meaning of robot's compliance, the proposed approach significantly reduced the force overshoot. This, in turn, minimizes the impact of the robot on the environment, thereby

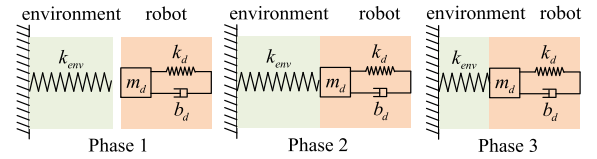


Fig. 1. The models of the environment and the robot in contact force control. Phase 1: without contact. Phase 2: contact occurs but $f_e = 0$. Phase 3: contact occurs with $f_e \neq 0$.

enhancing the safety of the robot's compliant operations. In operations such as grinding and polishing, overshoot in force can be avoided to prevent damaging the components. The main contributions are given as follows.

- 1) A new adaptive law is designed, which can provide a feed-forward compensation for uncertainties and significantly reduce the force overshoot during the contact transition compared with existing methods, further to improve force tracking accuracy.
- 2) A PD control has been built into the compensator combining with adaptive law, it can accelerate force response speed and further mitigate force overshoot.
- 3) The stability and convergence analysis is performed using a spring model, which confirms the rigor and effectiveness of the proposed theory. But it is not applicable to a complex second-order environmental model.

The rest parts of the letter are organized as follows. Section II introduces the fundamental theory of impedance control. Section III gives the design method of the new adaptive law and compensator. The stability and convergence analysis is performed. Simulations and experiments are presented in Section IV. Conclusions are given in Section V.

II. PRELIMINARIES

In this section, the environmental and robot model are introduced. Then the position-based impedance control and traditional adaptive impedance control have been discussed.

A. Environmental and Robot Model

In impedance control, the environment is presented by a second order mass-spring-damper system in some literatures. But there are also many works describe the environment model as a spring system, which is widely used in impedance control schemes and can be found in works [3], [15], [16], [18], [19], [20], [22]. For the rigid environment, a spring system model can represent most properties of it and is more convenient for controller design. Modeling errors are allowed during the modeling process, errors introduced by the environment model can be compensated by control law and adaptive law. This work focuses on rigid environments, so it is suitable to model the environment as a spring system with stiffness K_{env} . The robot is modeled as a second order mass-spring-damper system, with M_d , B_d , K_d the mass, damping and stiffness of robot end-effector. The contact process of robot and environment is given in Fig. 1. In Phase 1, robot approaches toward the environment without contact. In phase 2, robot contacts with environment but the contact force

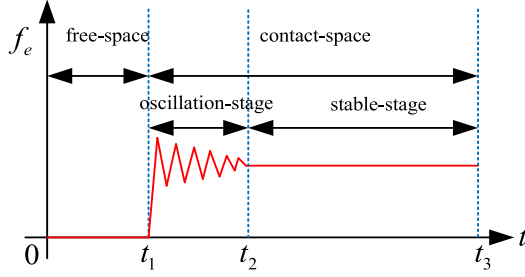


Fig. 2. Contact force variation in contact process.

is zero. In Phase 3, robot and environment achieve a relative equilibrium state, the contact force is not zero.

The contact force variation is shown in Fig. 2. In $0 - t_1$, robot approaches to the environment. In $t_1 - t_2$, robot contacts with environment and the contact force is oscillating and gradually decaying to the desired value. In $t_2 - t_3$, the contact force reaches a stable value.

B. Constant Impedance Control for Contact Force Tracking

In this section, we studied the position-based impedance control for contact force tracking and analyzed the force tracking error in various environment conditions.

Define X_r as reference position, X_c as command position, and X_m as actual position of robot end-effector. In the position control mode, it is regarded $X_c = X_m$ holds [16], [19]. Let $E_x = X_m - X_r = X_c - X_r$ represent position perturbation caused by contact force F_e , $F_e = K_{env}(X_e - X_m) = K_{env}(X_e - X_c)$. X_e and K_{env} are the position and stiffness of environment. Denote F_d as desired force between environment and robot, then the force tracking error is expressed as $\Delta F_e = F_e - F_d$.

The impedance function can be represented by a second order dynamical relation between position perturbation E_x and force error ΔF_e , it has

$$M_d \ddot{E}_x + B_d \dot{E}_x + K_d E_x = \Delta F_e \quad (1)$$

The transfer-function of system (1) is

$$H(s) = \frac{E_x(s)}{\Delta F_e(s)} = \frac{1}{M_d s^2 + B_d s + K_d} \quad (2)$$

The perturbation E_x of position is employed to adjust the required reference position X_r , further to generate command position X_c . It has $X_c = X_r + E_x = X_r + H(s) \cdot \Delta F_e(s)$. Then the robot controller tracks the command position X_c to achieve the desired contact force.

Next, we analyze the force tracking error ΔF_e with model (1). Given that each Cartesian variable is independent to others, we can, without loss of generality, examine it in one dimension. In this scenario, the force error is represented as

$$\begin{aligned} \Delta f_e &= f_e - f_d \\ &= k_{env}(x_e - x_c) - f_d \\ &= k_{env}x_e - k_{env}(x_r + h(s) \cdot \Delta f_e) - f_d \end{aligned} \quad (3)$$

Substitute $h(s) = \frac{1}{m_d s^2 + b_d s + k_d}$ into (3) yields

$$\Delta f_e = \frac{m_d s^2 + b_d s + k_d}{m_d s^2 + b_d s + k_d + k_{env}} [k_{env}(x_e - x_r) - f_d] \quad (4)$$

The steady-state force error can be derived from (4) as

$$\Delta f_{ss} = \frac{k_d}{k_d + k_{env}} [k_{env}(x_e - x_r) - f_d] \quad (5)$$

According to (5), the reference position must be set as

$$x_r = x_e - \frac{f_d}{k_{env}} \quad (6)$$

to eliminate the force tracking error.

If the environmental position x_e and stiffness k_{env} are available, they can be used in (6) to calculate reference position x_r . Then according to the impedance model (1), position perturbation e_x and command position x_c can be computed. But in practical applications, the environmental position x_e and stiffness k_{env} are either partially known or completely unknown, which results in the presence of force tracking error in conventional constant impedance control.

C. Traditional Adaptive Impedance Control

To cope with unknown environment, this section introduces the traditional adaptive impedance control scheme. Substitute $e_x = x_c - x_r$ and $\Delta f_e = f_e - f_d$ into (1), it yields

$$m_d(\ddot{x}_c - \ddot{x}_r) + b_d(\dot{x}_c - \dot{x}_r) + k_d(x_c - x_r) = f_e - f_d \quad (7)$$

Based on the analysis of (6), it is challenging to set reference position x_r due to unknown environmental geometry and stiffness. Works [16], [18], [19] proposed a new solution by replacing x_r with environmental position x_e in (7). We followed their works and obtained a new impedance relationship

$$m_d(\ddot{x}_c - \ddot{x}_e) + b_d(\dot{x}_c - \dot{x}_e) + k_d(x_c - x_e) = f_e - f_d \quad (8)$$

As noted in [18], setting the stiffness k_d to zero ensures the ideal steady-state condition $f_e = f_d$, regardless of the environmental stiffness k_{env} . Then (8) can be converted into

$$m_d(\ddot{x}_c - \ddot{x}_e) + b_d(\dot{x}_c - \dot{x}_e) = f_e - f_d \quad (9)$$

Note that in (9), if the environment is not a plane, which means $\dot{x}_e, \ddot{x}_e \neq 0$. The force tracking error always exists due to the actual values of $\dot{x}_e$ and $\ddot{x}_e$ are unknown.

Considering available environmental position as $\hat{x}_e = x_e - \delta x_e$, δx_e represents uncertainties of environmental position. Let $e_x = x_c - x_e$, then it has $\hat{e}_x = x_c - \hat{x}_e = e_x + \delta x_e$. Replacing e_x with $\hat{e}_x$ in (9), it derives

$$f_e - f_d = m_d \ddot{\hat{e}}_x + b_d \dot{\hat{e}}_x = m_d(\ddot{e}_x + \delta \ddot{x}_e) + b_d(\dot{e}_x + \delta \dot{x}_e) \quad (10)$$

To compensate environmental uncertainties, traditional adaptive impedance control law is designed as follows to adjust damping parameters

$$f_e - f_d = m_d \ddot{\hat{e}}_x + (b_d + \Delta b_d) \dot{\hat{e}}_x \quad (11)$$

Δb_d is designed as follows [19]

$$\begin{cases} \Delta b_d = \frac{b_d}{\hat{e}_x} \Psi(t) \\ \Psi(t) = \Psi(t - \mu) + \rho \frac{f_d(t - \mu) - f_e(t - \mu)}{b_d} \end{cases} \quad (12)$$

where μ represents the system's control period and ρ denotes the update rate.

The traditional adaptive impedance model (11)-(12) can cope with unknown environment, force tracking error can be minimized although the environmental position is unknown. However, large force overshoot will be caused when the robot contacts with environment. It is the main drawback in TAIPC.

III. THE PROPOSED IMPEDANCE CONTROL WITH ADAPTIVE COMPENSATOR

In this section, we introduce a novel adaptive law aimed at minimizing force tracking error and mitigating force overshoot in unknown environments.

A. Design of the Proposed Adaptive Impedance Control

Rewrite impedance model (9) as

$$m_d \ddot{x}_c + b_d \dot{x}_c + (f_d - f_e) = m_d \ddot{x}_e + b_d \dot{x}_e \quad (13)$$

If $\ddot{x}_e$ and $\dot{x}_e$ are available in (13), it can be used to generate command position x_c . But $\ddot{x}_e$ and $\dot{x}_e$ are unknown in practical applications. TAIPC (11)-(12) can deal with this problem and minimize force tracking error. However, large force overshoot will be caused in contact phase. In this work, we proposed a new adaptive compensator to cope with this problem.

Let $\tilde{f} = f_d - f_e$, then the following impedance model has been proposed

$$m_d \ddot{x}_c + b_d \dot{x}_c + \tilde{f} = u \quad (14)$$

where u is the compensator to deal with environmental uncertainties. It is formulated by two terms, first is an adaptive law and second is a PD control element, given as

$$\begin{cases} u = -d + \tilde{f} - \underbrace{\alpha \tilde{f} - \gamma \dot{\tilde{f}}}_{(I)} \\ \dot{d} = \underbrace{\beta(-u + \tilde{f} - d)}_{(II)} \end{cases} \quad (15)$$

The control structure of (14)-(15) is shown in Fig. 3. Note that impedance model (14) and (15) does not include environmental information. Therefore, the proposed approach is capable of handling unknown environments.

The design idea of control law (15) is described as follows. Note that a controller u is designed to cope with uncertainties in system (14), which is composed of two parts: A PD control element (I) and an additional term d produced by an adaptive compensator (II). The PD control can handle uncertainties, like used in works [8], [15], [22]. But only a PD term can not achieve a high-precision force tracking and a small force overshoot during contact transition. In this work, we introduced an additional term d in control law u to further compensate for the uncertainties. The term d is produced by an adaptive law (15) (II), which

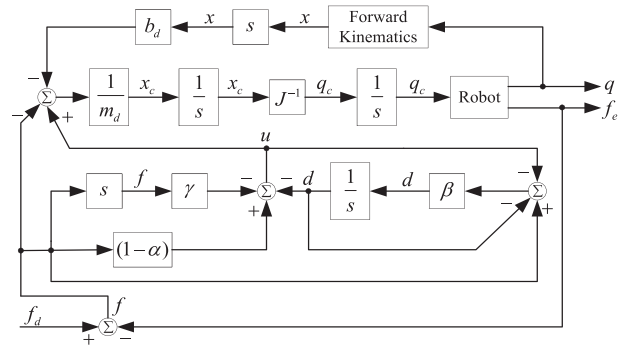


Fig. 3. The control structure of the proposed adaptive impedance control.

use the information of force tracking error $\tilde{f}$ and control u to update itself. The adaptive law (15) (II) serves as a feedforward compensation mechanism, it can adaptively estimate uncertainties and compensate them in control law. Therefore, the newly introduced adaptive law, compared to mere PD control, can enhance tracking accuracy and reduce force overshoot. More importantly, the adaptive law (15) (II) can provide a designable initial feedforward compensation $d(t_0)$, which offers effective compensation for uncertainties during the contact transition. This objective can not achieve with PD control and existing adaptive impedance control methods [18], [19], [20], [21], [22]. Therefore, the designed adaptive law can significantly reduce force overshoot.

The new adaptive compensator (15) is effective, brief and easy to implement. There are four variables needing to be set, gains α, β, γ and initial state of adaptive law $d(t_0)$. α can accelerate the force response speed. β can minimize the force tracking error in steady-state phase. γ is the gain of D control, it is well known that D control can reduce system overshoot in PID controller. $d(t_0)$ is the initial state of adaptive law, it affects the degree of compensation for environment uncertainties in initial phase. A suitable value of $d(t_0)$ can maximize the compensation for environment uncertainties.

B. The Stability and Convergence Analysis

This section presents the stability and convergence analysis of the proposed adaptive impedance control scheme (14)-(15) to validate the theoretical claims.

According to $f_e = k_{env}(x_e - x_c)$, it has $x_c = x_e - \frac{f_e}{k_{env}}$. Substituting it into (14) yields

$$m_d \left(\ddot{x}_e - \frac{\ddot{f}_e}{k_{env}} \right) + b_d \left(\dot{x}_e - \frac{\dot{f}_e}{k_{env}} \right) + \tilde{f} = u \quad (16)$$

$$- \frac{m_d}{k_{env}} \ddot{f}_e - \frac{b_d}{k_{env}} \dot{f}_e + \tilde{f} = u - m_d \ddot{x}_e - b_d \dot{x}_e \quad (17)$$

Adding $\frac{m_d}{k_{env}} \ddot{f}_d + \frac{b_d}{k_{env}} \dot{f}_d$ to both sides of (17) yields

$$\begin{aligned} & \frac{m_d}{k_{env}} (\ddot{f}_d - \ddot{f}_e) + \frac{b_d}{k_{env}} (\dot{f}_d - \dot{f}_e) + \tilde{f} \\ &= u + \frac{m_d}{k_{env}} (\ddot{f}_d - k_{env} \ddot{x}_e) + \frac{b_d}{k_{env}} (\dot{f}_d - k_{env} \dot{x}_e) \end{aligned} \quad (18)$$

Let $v = f_d - k_{env}x_e$, $m = \frac{m_d}{k_{env}}$, $b = \frac{b_d}{k_{env}}$, rewrite (18) as

$$m\ddot{f} + b\dot{f} + \tilde{f} = u + m\ddot{v} + b\dot{v} \quad (19)$$

The Laplace transform of (19) is

$$(ms^2 + bs + 1)\tilde{f}(s) = u(s) + (ms^2 + bs)v(s) \quad (20)$$

Performing the Laplace transform of (15) yields

$$u(s) = -d(s) + \tilde{f}(s) - \alpha\tilde{f}(s) - \gamma s\tilde{f}(s) \quad (21)$$

$$sd(s) = \beta(-u(s) + \tilde{f}(s) - d(s)) \quad (22)$$

According to (22), it has

$$d(s) = \frac{\beta}{s + \beta}(\tilde{f}(s) - u(s)) \quad (23)$$

Substituting (23) into (21) and simplifying, it yields

$$u(s) = \left(1 - \alpha - \gamma\beta - \gamma s - \frac{\alpha\beta}{s}\right)\tilde{f}(s) \quad (24)$$

Substituting (24) into (20) and simplifying, we can obtain

$$\left(ms^2 + (b + \gamma)s + (\alpha + \gamma\beta) + \frac{\alpha\beta}{s}\right)\tilde{f}(s) = (ms^2 + bs)v(s) \quad (25)$$

The transfer-function of system (25) is

$$g(s) = \frac{\tilde{f}(s)}{v(s)} = \frac{ms^3 + bs^2}{ms^3 + (b + \gamma)s^2 + (\alpha + \gamma\beta)s + \alpha\beta} \quad (26)$$

The stability and convergence of (26) are ensured by the characteristic (27)

$$ms^3 + (b + \gamma)s^2 + (\alpha + \gamma\beta)s + \alpha\beta = 0 \quad (27)$$

According to the Routh criterion, the Routh array is established as follows

$$\begin{array}{ccc} s^3 & m & (\alpha + \gamma\beta) \\ s^2 & (b + \gamma) & \alpha\beta \\ s^1 & \frac{1}{b + \gamma}[(\alpha + \gamma\beta)\gamma - \frac{1}{k_{env}}(m_d\alpha\beta - b_d(\alpha + \gamma\beta))] & 0 \\ s^0 & \alpha\beta & 0 \end{array} \quad (28)$$

For the system to be stable, the coefficients in the first column of the Routh array must be positive, signifying that

$$\begin{cases} m > 0 \\ b + \gamma > 0 \\ \frac{1}{b + \gamma}[(\alpha + \gamma\beta)\gamma - \frac{1}{k_{env}}(m_d\alpha\beta - b_d(\alpha + \gamma\beta))] > 0 \\ \alpha\beta > 0 \end{cases} \quad (29)$$

We can design α, β, γ to satisfy the relation

$$\begin{cases} \alpha, \beta, \gamma > 0 \\ m_d\alpha\beta < b_d(\alpha + \gamma\beta) \end{cases} \quad (30)$$

to ensure that (29) always holds. (30) is a sufficient condition of (29), which guarantees the system is stable.

The steady-state error can be expressed as

$$e_{ss} = \lim_{s \rightarrow 0} s(\tilde{f}(s) - v(s))$$

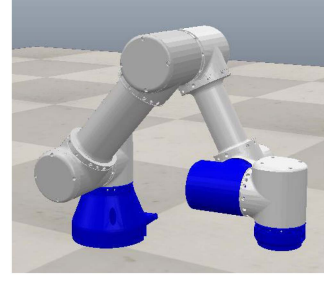


Fig. 4. Zerror6 robot simulation model.

TABLE I
MODIFIED DH PARAMETERS OF ZERROR6 ROBOT

	a_i	α_i	d_i	θ_i
1	0	0	0.1435	θ_1
2	0	$\frac{\pi}{2}$	0.098	θ_2
3	0.2435	0	0	θ_3
4	0.211	π	0.0115	θ_4
5	0	$-\frac{\pi}{2}$	-0.086	θ_5
6	0	$\frac{\pi}{2}$	-0.0725	θ_6

$$= \lim_{s \rightarrow 0} s \left[\frac{ms^3 + bs^2}{ms^3 + (b + \gamma)s^2 + (\alpha + \gamma\beta)s + \alpha\beta} v(s) - v(s) \right] \quad (31)$$

Consider the input as a step function, it has $v(s) = \frac{1}{s}$. Then (31) yields

$$e_{ss} = \lim_{s \rightarrow 0} s(\tilde{f}(s) - v(s)) = -1 \quad (32)$$

$$\lim_{s \rightarrow 0} s\tilde{f}(s) = 0 \quad (33)$$

$$\lim_{t \rightarrow \infty} \tilde{f}(t) = 0 \quad (34)$$

As a conclusion, it has $f_e \rightarrow f_d$ when $t \rightarrow \infty$. The contact force between the environment and robot can converge to the desired value. The stability of the force tracking control system has been guaranteed.

IV. SIMULATIONS AND EXPERIMENTS

To verify the performance of the proposed adaptive impedance controller, we construct a serious simulations and experiments in different scenes. Simulations are performed in three cases which covers most actual situations. Then three experiments are carried out in practical applications.

A. Simulations Analysis

The simulations are performed on a 6-Dof robot manipulator, which is shown in Fig. 4. Its modified DH parameters are given in Table I. The simulation studies are considered in three cases. Four methods are performed and compared: constant impedance control (CIPC) (9), traditional adaptive impedance control (TAIPC) (11)-(12), traditional adaptive impedance control with PD compensator (TAIPC-PD), proposed novel adaptive impedance control in this work (NAIPC).

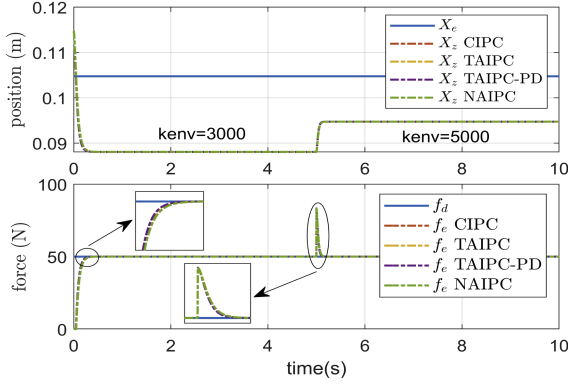


Fig. 5. Force tracking result on varying stiffness plane in simulation.

TABLE II

THE QUANTITATIVE COMPARISONS FOR FORCE TRACKING PERFORMANCE WITH DIFFERENT METHODS IN CASES 1-3 IN SIMULATION

	Plane surface		Slope surface		Sinusoidal surface	
	Overshoot (N)	Setting time(s)	Overshoot (N)	Setting time(s)	Overshoot (N)	Setting time(s)
CIPC	0	0.3	3.58	inf	5.53	inf
TAIPC	0	0.3	12.19	1.31	26.5	0.34
TAIPC-PD	0	0.28	10.67	1.32	21.8	0.37
NAIPC	0	0.29	0	0.71	0	0.33

Note: 'inf' means the setting time is infinite and the algorithm does not converge.

1) *Force Tracking on Varying Stiffness Plane:* In this case, the robot operates on a plane surface with varying stiffness, and the environmental stiffness is defined as

$$k_{env} = \begin{cases} 3000, & 0 \leq t \leq 5s \\ 5000, & 5 < t \leq 10s \end{cases} \quad (35)$$

The impedance parameters are chosen to ensure optimal force tracking performance for each approach by the trial-and-error method, CIPC ($m_d = 2, b_d = 200$), TAIPC ($m_d = 2, b_d = 200, \rho = 1.0 \times 10^{-5}$), TAIPC-PD ($m_d = 2, b_d = 200, \rho = 1.0 \times 10^{-5}, k_p = 0.1, k_d = 0.001$), NAIPC ($m_d = 2, b_d = 200, \alpha = 1, \beta = 1.0 \times 10^{-5}, \gamma = 0.001, d(0) = 0$). The desired force $f_d = 50N$. The force tracking result is presented in Fig. 5. The environmental stiffness changes from 3000 to 5000 at 5s, actual force f_e can still track desired force f_d after a brief force adjustment. The quantitative analysis is given in columns 2 and 3 in Table II. It is note that the overshoot of different methods is 0. The setting time is almost the same. The result illustrates that the proposed approach can work just as well as others on a plane. The newly introduced adaptive compensator (15) does not destroy the performance of the typical impedance control framework.

2) *Force Tracking on a Slope Surface:* In this case, the robot works on a slope surface, the environmental stiffness is set as $k_{env} = 5000$. The impedance parameters m_d, b_d, f_d are same as in Case 1. Other parameters are set as TAIPC ($\rho = 0.003$), TAIPC-PD ($\rho = 0.003, k_p = 0.1, k_d = 0.01$), NAIPC ($\alpha = 1, \beta = 5, \gamma = 0.01, d(0) = -50$). The force tracking result is presented in Fig. 6. Obviously that method CIPC can not track desired force on a slope surface, it has a large steady-state

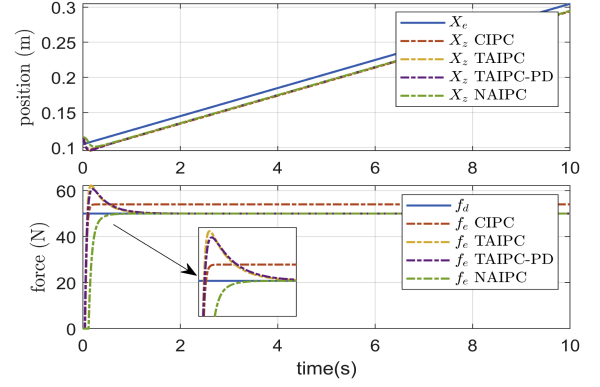


Fig. 6. Force tracking result on slope surface in simulation.

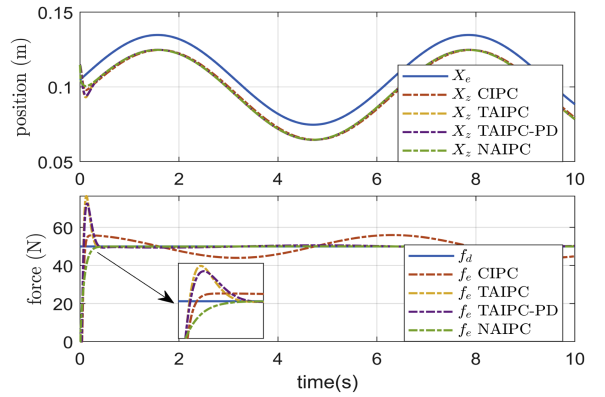


Fig. 7. Force tracking result on sinusoidal surface in simulation.

force error, which is caused by unknown environmental position ($\dot{x}_e \neq 0$). Methods TAIPC and TAIPC-PD can track desired force after a brief force adjustment, but large force overshoot exists in contact phase. Method NAIPC can track desired force without force overshoot and has faster force response speed. The quantitative analysis is given in columns 4 and 5 in Table II. Note that method NAIPC has smaller overshoot and setting time compared with others. The result demonstrates that the proposed approach can cope with unknown environment effectively and reduce force overshoot significantly.

3) *Force Tracking on a Sinusoidal Surface:* In this case, the robot works on a sinusoidal surface, the environmental stiffness is set as $k_{env} = 5000$. The impedance parameters m_d, b_d, f_d are same as in Case 1. Other parameters are designed as TAIPC ($\rho = 0.01$), TAIPC-PD ($\rho = 0.01, k_p = 0.1, k_d = 0.01$), NAIPC ($\alpha = 6, \beta = 15, \gamma = 0.5, d(0) = 0$). The force tracking result is presented in Fig. 7. Note that method CIPC can not track desired force on a sinusoidal surface, which is due to unknown environmental position ($\dot{x}_e \neq 0, \ddot{x}_e \neq 0$). Methods TAIPC and TAIPC-PD can track desired force, but they have large force overshoot. Method NAIPC can track desired force well without force overshoot. The quantitative analysis is given in columns 6 and 7 in Table II. Note that the overshoot of NAIPC is 0, but large for other methods (5.53N, 26.5N, 21.8N). The result sufficiently verifies the better force overshoot reduction ability of the proposed method NAIPC.

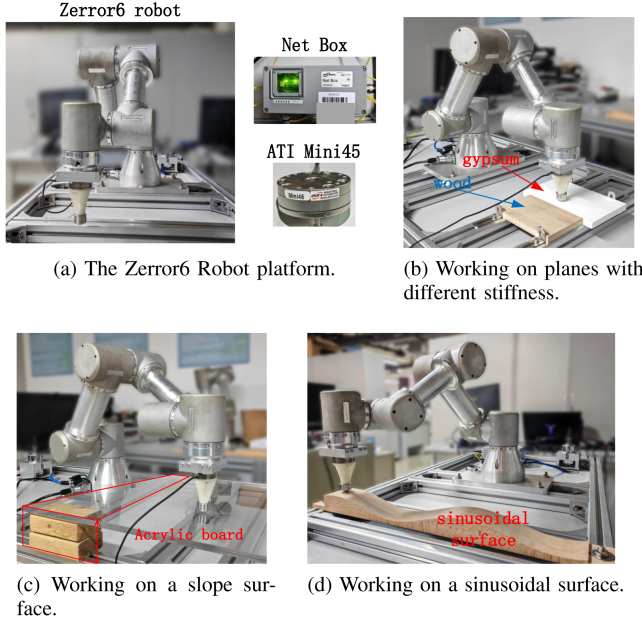


Fig. 8. Experiment platform and different working scenes.

B. Experiment Studies

The experiments are implemented on a 6-Dof robot platform (Fig. 8(a)), which works in a position mode. The controller of robot is an Industrial Personal Computer (IPC) with operation system Ubuntu22.04, equipped with RT kernel Linux-5.10.180-rt88. The algorithm is implemented with C++ code, with 1 ms control period based on the EtherCAT bus. The 6D force sensor ATI Mini45 is equipped on the robot end-effector to measure contact force, which communicate with robot controller via TCP/IP, with communication cycle 1 ms. Also, four methods CIPC, TAIPC, TAIPC-PD and NAIPC are performed in different scenes and analyzed.

1) *Force Tracking on Varying Stiffness Planes*: The force tracking is performed on two planes with different stiffness, gypsum plane and wood plane, which are pieced together and presented in Fig. 8(b). The speed is 0.015 m/s in robot moving direction. The selection rules of impedance parameters are same as in simulation, CIPC ($m_d = 26, b_d = 2500$), TAIPC ($m_d = 26, b_d = 2500, \rho = 1.0 \times 10^{-5}$), TAIPC-PD ($m_d = 26, b_d = 2500, \rho = 1.0 \times 10^{-5}, k_p = 0.01, k_d = 0.02$), NAIPC ($m_d = 26, b_d = 2500, \alpha = 1, \beta = 1.0 \times 10^{-5}, \gamma = 0.02, d(0) = 0$). The desired force $f_d = 20$ N. The force tracking result is shown in Fig. 9. Note that when the robot contact with plane, methods CIPC and TAIPC have large force overshoot. Methods TAIPC-PD and NAIPC have little force overshoot. After the stiffness changed, four methods can track desired force after a brief force adjustment. The quantitative analysis is given in columns 2 and 3 in Table III. Note that both force overshoot (0.46N) and setting time (1.81 s) are smaller of method NAIPC compared with others. The result illustrates that method NAIPC can work just as well as other methods on a stiffness changing plane. The newly introduced adaptive compensator (15) does not destroy the performance of the typical impedance controller.

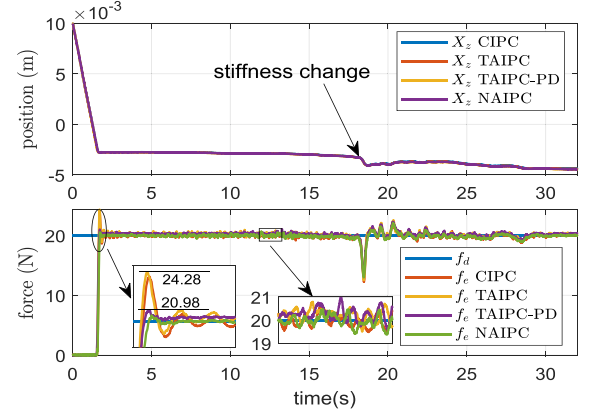


Fig. 9. Force tracking result on varying stiffness planes in experiment.

TABLE III
THE QUANTITATIVE COMPARISONS FOR FORCE TRACKING PERFORMANCE WITH DIFFERENT METHODS IN SCENES 1-3 IN EXPERIMENT

	Plane surface		Slope surface		Sinusoidal surface	
	Overshoot (N)	Setting time(s)	Overshoot (N)	Setting time(s)	Overshoot (N)	Setting time(s)
CIPC	3.84	3.01	0.89	inf	1.36	inf
TAIPC	4.28	3.02	41.64	1.98	57.1	1.39
TAIPC-PD	0.98	1.83	21.33	2.94	22.8	2.01
NAIPC	0.46	1.81	0.12	1.2	2.58	0.68

Note: 'inf' means the setting time is infinite and the algorithm does not converge.

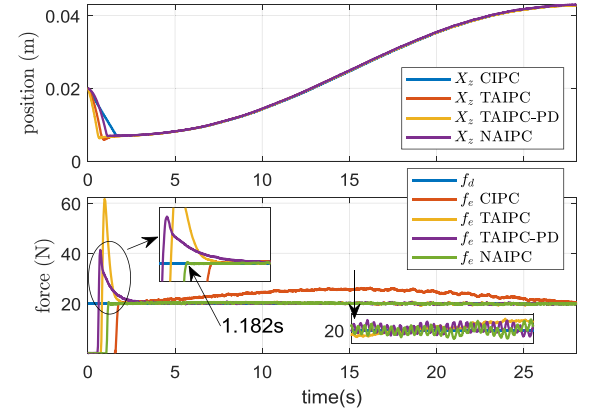


Fig. 10. Force tracking result on a slope surface in experiment.

2) *Force Tracking on a Slope Surface*: In this scene, the robot works on a slope surface with acrylic material (Fig. 8(c)), with speed 0.013 m/s in robot moving direction. The impedance parameters m_d, b_d, f_d are same as in Scene 1. Other parameters are set as TAIPC ($\rho = 0.003$), TAIPC-PD ($\rho = 0.003, k_p = 1, k_d = 0.01$), NAIPC ($\alpha = 2.8, \beta = 0.8, \gamma = 0.056, d(0) = -50$). The force tracking result is presented in Fig. 10. Note that position trajectory is not a strict slope, this is because the acrylic board undergoes deformation during contact process. It can be seen that method CIPC can not track desired force, it has a large force error, which is caused by unknown environmental information ($\ddot{x}_e \neq 0$ and $\dot{x}_e \neq 0$). Methods TAIPC and TAIPC-PD can track desired force, but large force overshoot exists in contact phase.

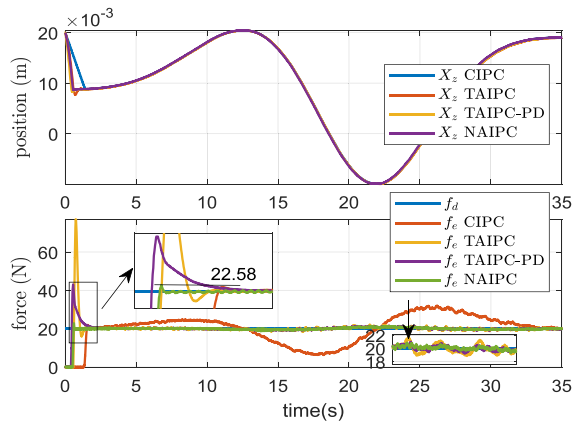


Fig. 11. Force tracking result on a sinusoidal surface in experiment.

The quantitative analysis is given in columns 4 and 5 in Table III. The force overshoot of TAIPC and TAIPC-PD are 41.64N and 21.33N. The force overshoot of NAIPC is 0.12 N, it has been significantly reduced. The setting time (1.2s) of NAIPC is also smaller than TAIPC and TAIPC-PD. As a conclusion, the proposed approach has better force overshoot reduction ability and faster response speed compared with others.

3) *Force Tracking on a Sinusoidal Surface*: In this scene, the robot works on a sinusoidal surface with wood material (Fig. 8(d)). The speed in robot moving direction is 0.012 m/s. The impedance parameters m_d , b_d , f_d are same as in Case 1. Other parameters are set as TAIPC ($\rho = 0.005$), TAIPC-PD ($\rho = 0.005$, $k_p = 1.3$, $k_d = 0.02$), NAIPC ($\alpha = 4.2$, $\beta = 1.3$, $\gamma = 0.06$, $d(0) = -60$). The force tracking result is presented in Fig. 11. Note that method CIPC can not track desired force on a sinusoidal surface. Methods TAIPC and TAIPC-PD can track desired force, but with large force overshoot. Method NAIPC can track desired force on a sinusoidal surface with little force overshoot and smaller setting time. The quantitative analysis is given in columns 6 and 7 in Table III. Note that force overshoot of NAIPC is 2.58 N, of TAIPC and TAIPC-PD are 57.1 N and 22.8 N. The overshoot has been reduction a lot by method NAIPC. The setting time (0.68 s) of NAIPC is also smaller than other approaches.

V. CONCLUSION

In this work, we proposed a novel robust adaptive impedance control scheme for contact force tracking with unknown environment. A new brief and effective adaptive law has been designed to minimize the force tracking error under unknown environmental stiffness and geometry. Then a PD control element is employed to construct a compensator combining with the adaptive law, to further accelerate force response speed and alleviate force overshoot. Compared with existed methods, CIPC, TAIPC and TAIPC-PD, the proposed method NAIPC can not only track desired force with good precision, but also reduce force overshoot significantly. It is robust to unknown environment. The stability analysis with a spring model confirms its theoretical claims, but the method is not suitable for a complex second-order environmental model. Abundant simulations and experiments in

different working scenarios demonstrate the better force tracking performance of the proposed scheme.

REFERENCES

- [1] P. Maurice, M. E. Huber, N. Hogan, and D. Sternad, "Velocity-curvature patterns limit human-robot physical interaction," *IEEE Robot. Autom. Lett.*, vol. 3, no. 1, pp. 249–256, Jan. 2018.
- [2] X. Yu, S. Liu, S. Zhang, W. He, and H. Huang, "Adaptive neural network force tracking control of flexible joint robot with an uncertain environment," *IEEE Trans. Ind. Electron.*, vol. 71, no. 6, pp. 5941–5949, Jun. 2024.
- [3] S. Jung and T. Hsia, "Robust neural force control scheme under uncertainties in robot dynamics and unknown environment," *IEEE Trans. Ind. Electron.*, vol. 47, no. 2, pp. 403–412, Apr. 2000.
- [4] M. H. Raibert and J. J. Craig, "Hybrid position/force control of manipulators," *J. Dynamic Syst., Meas., Control*, vol. 103, no. 2, pp. 126–133, 1981.
- [5] N. Hogan, "Impedance control: An approach to manipulation," in *Proc. 1984 Amer. Control Conf.*, 1984, pp. 304–313.
- [6] S. Ding, J. Peng, J. Xin, H. Zhang, and Y. Wang, "Task-oriented adaptive position/force control for robotic systems under hybrid constraints," *IEEE Trans. Ind. Electron.*, vol. 71, no. 10, pp. 12612–12622, Oct. 2024.
- [7] M. Abbas, S. Al Issa, and S. K. Dwivedy, "Event-triggered adaptive hybrid position-force control for robot-assisted ultrasonic examination system," *J. Intell. Robot. Syst.*, vol. 102, no. 4, 2021, Art. no. 84.
- [8] L. Gu and Q. Huang, "Adaptive impedance control for force tracking in manipulators based on fractional-order pid," *Appl. Sci.-Basel*, vol. 13, no. 18, 2023, Art. no. 10267.
- [9] B. Zhou, F. Song, Y. Liu, F. Fang, and Y. Gan, "Robust sliding mode impedance control of manipulators for complex force-controlled operations," *Nonlinear Dyn.*, vol. 111, no. 24, pp. 22267–22281, 2023.
- [10] Z. Jin, D. Qin, A. Liu, W.-A. Zhang, and L. Yu, "Model predictive variable impedance control of manipulators for adaptive precision-compliance tradeoff," *IEEE-ASME Trans. Mechatron.*, vol. 28, no. 2, pp. 1174–1186, Apr. 2023.
- [11] T. Sun, Z. Wang, C. He, and L. Yang, "Adaptive robust admittance control of robots using duality principle-based impedance selection," *Appl. Sci.-Basel*, vol. 12, no. 23, 2022, Art. no. 12222.
- [12] G. A. Medrano-Cerda, "Counterexample to passivity preservation for variable impedance control of compliant robots," *IEEE/ASME Trans. Mechatron.*, vol. 28, no. 1, pp. 598–602, Feb. 2023.
- [13] M. Sharifi, A. Zakerimanesh, J. K. Mehr, A. Torabi, V. K. Mushahwar, and M. Tavakoli, "Impedance variation and learning strategies in human-robot interaction," *IEEE Trans. Cybern.*, vol. 52, no. 7, pp. 6462–6475, Jul. 2022.
- [14] E. Spyarakos-Papastavridis, P. R. N. Childs, and J. S. Dai, "Passivity preservation for variable impedance control of compliant robots," *IEEE-ASME Trans. Mechatron.*, vol. 25, no. 5, pp. 2342–2353, Oct. 2020.
- [15] Z. Liu and Y. Sun, "Adaptive variable impedance control with fuzzy-pi compound controller for robot trimming system," *Arabian J. Sci. Eng.*, vol. 47, no. 12, pp. 15727–15740, 2022.
- [16] M. H. Hamedani, H. Sadeghian, M. Zekri, F. Sheikholeslam, and M. Keshmiri, "Intelligent impedance control using wavelet neural network for dynamic contact force tracking in unknown varying environments," *Control Eng. Pract.*, vol. 113, 2021, Art. no. 104840.
- [17] Z. Jin, A. Liu, W.-a. Zhang, and L. Yu, "An optimal variable impedance control with consideration of the stability," *IEEE Robot. Autom. Lett.*, vol. 7, no. 2, pp. 1737–1744, Apr. 2022.
- [18] S. Jung, T. Hsia, and R. Bonitz, "Force tracking impedance control of robot manipulators under unknown environment," *IEEE Trans. Control Syst. Technol.*, vol. 12, no. 3, pp. 474–483, May 2004.
- [19] D. Jinjun, G. Yahui, C. Ming, and D. Xianzhong, "Adaptive variable impedance control for dynamic contact force tracking in uncertain environment," *Robot. Auton. Syst.*, vol. 102, pp. 54–65, 2018.
- [20] J. Dong, J. Xu, Q. Zhou, and S. Hu, "Physical human-robot interaction force control method based on adaptive variable impedance," *J. Franklin Inst.-Eng. Appl. Math.*, vol. 357, no. 12, pp. 7864–7878, 2020.
- [21] B. Dong, Y. Jing, X. Zhu, Y. Cui, and T. An, "Adaptive impedance decentralized control of modular robot manipulators for physical human-robot interaction," *J. Intell. Robot. Syst.*, vol. 109, no. 3, 2023, Art. no. 48.
- [22] G. Wang, Y. Deng, H. Zhou, and X. Yue, "PD-adaptive variable impedance constant force control of macro-mini robot for compliant grinding and polishing," *Int. J. Adv. Manuf. Technol.*, vol. 124, no. 7–8, pp. 2149–2170, 2023.